

Imagine! Invent!

*"To invent, you need a good imagination
and a pile of junk."*

— THOMAS EDISON

ESSENTIAL QUESTION

What does it take to make a successful invention?

MAIN SELECTIONS

1 **Timeless Thomas** · Gene Barretta

BIOGRAPHY

2 **A Bumpy Ride** · Sharon Katz Cooper & Rachel Young

INFORMATIONAL

3 **Rosie Revere, Engineer** · Andrea Beaty

NARRATIVE POEM

4 **Edison's Best Invention**

OPINION TEXT

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B · Reading Comprehension (32)	p. 6
C · IB Critical Thinking (5)	p. 13
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Worksheet A — Vocabulary

Module 8 · Imagine! Invent! · Grade 3 · 26 Items

CCSS L.3.4 · L.3.5 · L.3.6

NAME _____ DATE _____

Directions. Read each question carefully. Watch out for tricky choices. Fill in the circle next to the BEST answer.

BIG IDEA WORDS

Q1 L.3.6

invention

An **invention** is —

- ☐ **A.** a copy of something that already exists.
- ☐ **B.** a story about people who are not real.
- ☐ **C.** something a person creates that did not exist before.
- ☐ **D.** a rule that everyone has to follow.

Q2 L.3.6

brilliant

When an idea or a person is **brilliant**, it is —

- ☐ **A.** extremely clever or skillful.
- ☐ **B.** dull and not very useful.
- ☐ **C.** heavy and hard to lift.
- ☐ **D.** old and out of date.

Q3 L.3.5 **TRAP**

productive

If you are **productive**, you —

- ☐ **A.** waste most of your time.
- ☐ **B.** get tired before you begin.
- ☐ **C.** do the same thing over and over by mistake.
- ☐ **D.** are able to do a lot with the time and resources you have.

Q4 L.3.6

original

Something that is **original** is —

- ☐ **A.** borrowed from a friend.
- ☐ **B.** the first of its kind.
- ☐ **C.** broken into many pieces.
- ☐ **D.** the very last one left.

CRITICAL VOCABULARY · TIMELESS THOMAS

Q5 L.3.4

valuable

Edison's failures were **valuable** to him. If something is **valuable**, it is —

- ☐ **A.** useful, helpful, or important.
- ☐ **B.** easy to throw away.
- ☐ **C.** made only of metal.
- ☐ **D.** too small to notice.

Q6 L.3.4

device

The phonograph was the first **device** to record sound. A **device** is —

- ☐ **A.** a kind of song.
- ☐ **B.** a person who does experiments.
- ☐ **C.** a tool or machine that has a certain purpose.
- ☐ **D.** a place where people work.

Q7 L.3.5 **TRAP**

breakthrough

The phonograph was a major **breakthrough**. If you make a **breakthrough**, you —

- ☐ **A.** break a window by accident.
- ☐ **B.** run all the way through a wall.
- ☐ **C.** take a short rest from your work.
- ☐ **D.** make an important discovery after many tries.

Q8 L.3.4

dictation

Office workers used the phonograph for **dictation**. **Dictation** is —

- ☐ **A.** the act of drawing a picture of a machine.
- ☐ **B.** the act of writing down or recording words that have been spoken.
- ☐ **C.** the act of fixing a broken tool.
- ☐ **D.** the act of selling something at a store.

CRITICAL VOCABULARY · TIMELESS THOMAS (CONTINUED)

Q9 L.3.4

technology

Edison's fluoroscope used X-ray **technology**. **Technology** is —

- ☐ **A.** the use of science to invent useful things or to solve problems.
- ☐ **B.** a story about the future.
- ☐ **C.** a kind of music played at night.
- ☐ **D.** a tool used only for farming.

Q10 L.3.4

radar

Edison was among the first to discover the radio waves used in **radar**. **Radar** is —

- ☐ **A.** a bright kind of electric light.
- ☐ **B.** a machine that records music.
- ☐ **C.** a way to find unseen objects by using radio signals.
- ☐ **D.** a hand tool for cutting metal.

CRITICAL VOCABULARY · A BUMPY RIDE

Q11 L.3.4

contraption

Karl von Drais built a **contraption** with no pedals. A **contraption** is —

- ☐ **A.** a wide, paved road.
- ☐ **B.** a kind of horse.
- ☐ **C.** a special riding school.
- ☐ **D.** a mechanical gadget or device that has a certain purpose.

Q12 L.3.4

quest

Inventors went on a **quest** to make bikes safer. If you go on a **quest**, you —

- ☐ **A.** go in search of something important to you.
- ☐ **B.** sit still and wait for help.
- ☐ **C.** ask a question on a test.
- ☐ **D.** rest after a long day.

Q13 L.3.5 **TRAP**

craze

In the 1890s, cycling became a national **craze**. If there is a **craze** for something, it is —

- ☐ **A.** broken into tiny cracks.
- ☐ **B.** wild and dangerous to be near.
- ☐ **C.** very popular for a period of time.
- ☐ **D.** hidden away where no one can see it.

Q14 L.3.4

era

Bikes helped end the **era** of long dresses. An **era** is —

- ☐ **A.** a mistake that gets fixed.
- ☐ **B.** a period of time in history.
- ☐ **C.** a kind of bicycle race.
- ☐ **D.** a tool for paving roads.

CRITICAL VOCABULARY · ROSIE REVERE, ENGINEER

Q15 L.3.4

engineer

Rosie dreamed of becoming a great **engineer**. An **engineer** is —

- ☐ **A.** a person who paints pictures for fun.
- ☐ **B.** a person who teaches a class.
- ☐ **C.** a person who sells things at a market.
- ☐ **D.** a person who designs, builds, or operates machines.

Q16 L.3.5 TRAP

perplexed

Rosie felt **perplexed** by her aunt's reaction. If you are **perplexed**, you feel —

- ☐ **A.** confused and worried about something.
- ☐ **B.** calm and perfectly sure.
- ☐ **C.** sleepy after a long nap.
- ☐ **D.** proud of a finished plan.

Q17 L.3.4

dynamo

Great-Great-Aunt Rose was a true **dynamo**. A person who is a **dynamo** —

- ☐ **A.** is always tired and slow.
- ☐ **B.** never tries anything new.
- ☐ **C.** has a lot of energy and is very good at what that person does.
- ☐ **D.** is afraid to leave the house.

Q18 L.3.5 TRAP

lingers

Aunt Rose said time never **lingers**. When something **lingers**, it —

- ☐ **A.** disappears right away.
- ☐ **B.** remains for a long time.
- ☐ **C.** moves in a fast circle.
- ☐ **D.** breaks into pieces.

Q19 L.3.5 TRAP

whirled

The cheese-copter **whirled** round and round. Something that **whirled** —

- ☐ **A.** made a loud whirring noise but stayed still.
- ☐ **B.** floated quietly without moving.
- ☐ **C.** fell straight down to the ground.
- ☐ **D.** turned around several times.

Q20 L.3.4

baffled

Young Rosie was **baffled**. If you are **baffled**, you are —

- ☐ **A.** confused.
- ☐ **B.** cheerful.
- ☐ **C.** sleepy.
- ☐ **D.** hungry.

CRITICAL VOCABULARY · EDISON'S BEST INVENTION

Q21 L.3.4

visionary

Edison was a **visionary**. A **visionary** is a person who —

- ☐ **A.** can see far away in the dark.
- ☐ **B.** only copies what others have done.
- ☐ **C.** has new or unusual ideas about life in the future.
- ☐ **D.** wears glasses to read small print.

Q22 L.3.5 TRAP

compact

A cell phone is small and **compact**. Things that are **compact** —

- ☐ **A.** can fit in a small space.
- ☐ **B.** are packed in a paper box.
- ☐ **C.** are very large and heavy.
- ☐ **D.** make a loud popping sound.

Q23 L.3.4

singles

Without recorded sound there would be no music **singles**. Music **singles** are —

- ☐ **A.** people who sing all alone.
- ☐ **B.** recordings of one song.
- ☐ **C.** the words written inside a book.
- ☐ **D.** large bands of many players.

Q24 L.3.4

memorable

The phonograph saved **memorable** speeches. When something is **memorable**, it is —

- ☐ **A.** easy to forget right away.
- ☐ **B.** too quiet to hear.
- ☐ **C.** written only in one language.
- ☐ **D.** special enough that people want to remember it.

Q25 L.3.4

system

Edison built a light and power **system**. A **system** is —

- ☐ **A.** a set of things or ideas that work together to get something done.
- ☐ **B.** a single light bulb in a room.
- ☐ **C.** a story about an inventor.
- ☐ **D.** a tool used to dig in the ground.

Q26 L.3.5 **TRAP**

innovative

Edison's **innovative** grid brought electricity into homes. An **innovative** idea is one that —

- ☐ **A.** is brand new in its packaging from a store.
- ☐ **B.** has never been thought of before.
- ☐ **C.** is copied straight from an old plan.
- ☐ **D.** works only inside one building.



Worksheet B — Reading Comprehension

Module 8 · Imagine! Invent! · Grade 3 · 32 Items

CCSS RI.3.1 · RI.3.2 · RI.3.3 · RI.3.5 · RI.3.6 · RI.3.8 · RL.3.1 · RL.3.2 · RL.3.3 · RL.3.5 · RL.3.6

NAME _____ DATE _____

Directions. Use what you remember from each selection to answer the questions. Fill in the circle next to the BEST answer.

Selection 1 · Timeless Thomas

BIOGRAPHY / INFORMATIONAL · GENE BARRETTA · 8 QUESTIONS

Q1 RI.3.2

What is this biography mostly about?

- ☐ A. How to build a phonograph at home.
- ☐ B. How Thomas Edison's inventions changed the way we live today.
- ☐ C. Why electric cars never became popular.
- ☐ D. How to work as a telegraph operator.

Q2 RI.3.3

How did Edison do so much inventing?

- ☐ A. He worked completely alone in his bedroom.
- ☐ B. He bought finished inventions from other people.
- ☐ C. He gathered a large team and started the world's first research laboratory, the Invention Factory.
- ☐ D. He stopped working whenever something failed.

Q3 RI.3.3

How did Edison treat his failures?

- ☐ A. He used them as a necessary part of inventing and always tried again.
- ☐ B. He gave up inventing after the first one.
- ☐ C. He hid them so no one would know.
- ☐ D. He blamed his team of scientists.

Q4 RI.3.5

The author puts "Present Day" pages next to "Edison's Lab" pages. What text structure is this?

- ☐ **A.** A list of cooking steps.
- ☐ **B.** A made-up fairy tale.
- ☐ **C.** A set of riddles to solve.
- ☐ **D.** Compare and contrast, showing how today's inventions relate to Edison's original ones.

Q5 RI.3.3

Why was Edison's tinfoil phonograph so important?

- ☐ **A.** It was the first device to record sound and play it back.
- ☐ **B.** It was the first electric car.
- ☐ **C.** It was the first camera ever made.
- ☐ **D.** It was the first battery for a toy.

Q6 RI.3.3

Edison first made his nickel-iron storage battery to power an electric car. What happened then?

- ☐ **A.** The battery never worked at all.
- ☐ **B.** The electric car didn't catch on, so he found many other uses for the battery.
- ☐ **C.** He threw the battery away and forgot it.
- ☐ **D.** He sold the battery to the Wright brothers.

Q7 RI.3.8

The author says Edison's lab was where the movie industry "essentially started." Which detail BEST supports this idea?

- ☐ **A.** Edison was partially deaf.
- ☐ **B.** Edison began as a telegraph operator.
- ☐ **C.** Edison created the first motion picture camera, the Kinetograph, and built the first movie studio.
- ☐ **D.** Edison had a pet bear cub at his lab.

Q8 RI.3.2 **TRAP**

What is the MAIN idea of "Timeless Thomas"?

- ☐ **A.** Edison was awarded 1,093 patents, and his very first one was for a vote recorder.
- ☐ **B.** Edison's movie studio was called the Black Maria because it looked like a police patrol wagon.
- ☐ **C.** Edison began experimenting when he was a boy, even though he was partially deaf.
- ☐ **D.** A curious inventor who never gave up changed the world with inventions we still use today.

Selection 2 · A Bumpy Ride

INFORMATIONAL · SHARON KATZ COOPER & RACHEL YOUNG / ARIANE ELSAMMAK · 8 QUESTIONS

Q9 RI.3.2

What is "A Bumpy Ride" mostly about?

- ☐ **A.** How the bicycle was invented and improved over many years.
- ☐ **B.** How to ride a bike without falling.
- ☐ **C.** How horses were used on dirt roads.
- ☐ **D.** How to fix a flat tire quickly.

Q10 RI.3.3

What was Karl von Drais's draisienne like?

- ☐ **A.** It had a chain and rubber tires.
- ☐ **B.** It had no pedals and was more like a sit-on scooter.
- ☐ **C.** It had a huge front wheel five feet across.
- ☐ **D.** It had a small motor on the back.

Q11 RI.3.3

Why was Kirkpatrick Macmillan's pedal-powered bicycle a problem?

- ☐ **A.** It could not move at all.
- ☐ **B.** It was made only for mail carriers.
- ☐ **C.** It cost almost nothing to buy.
- ☐ **D.** It was hard to steer, and he caused the first bike wreck.

Q12 RI.3.3

How did the air-filled tire come to be invented?

- ☐ **A.** Queen Victoria designed it for her trikes.
- ☐ **B.** Pierre Michaux added it to the front wheel.
- ☐ **C.** John Boyd Dunlop wrapped an air-filled hose around his son's tricycle wheels.
- ☐ **D.** Karl von Drais made it for his draisienne.

Q13 RI.3.5

The author uses a timeline with years like 1818, 1870, and 1889. How does this help the reader?

- ☐ **A.** It tells the price of each bicycle.
- ☐ **B.** It shows the important dates in the history of the bicycle, in order.
- ☐ **C.** It explains how to build a wheel.
- ☐ **D.** It names every rider who owned a bike.

Q14 RI.3.1

According to the selection, why did the "ordinaries" of 1870 have such a huge front wheel?

- ☐ **A.** A bigger wheel went farther with each turn of the pedals, so the bike could go faster.
- ☐ **B.** A big wheel made the bike cheaper to build.
- ☐ **C.** A big wheel kept the rider close to the ground.
- ☐ **D.** A big wheel was needed to hold the brakes.

Q15 RI.3.6

Why does the author give the article the title "A Bumpy Ride"?

- ☐ **A.** Because the article is only about rough dirt roads.
- ☐ **B.** Because every bicycle today still gives a bumpy ride.
- ☐ **C.** Because early bikes were bumpy AND the history had many bumps and problems along the way.
- ☐ **D.** Because the author did not enjoy riding bikes.

Q16 RI.3.2 **TRAP**

What is the MAIN idea of "A Bumpy Ride"?

- ☐ **A.** Queen Victoria of England bought two trikes, which had no brakes at all.
- ☐ **B.** Some early riders wore leather caps, but they did little to cushion a fall.
- ☐ **C.** The fast, big-wheeled bikes of 1870 were called "ordinaries."
- ☐ **D.** Many inventors each improved on the last, and together their ideas made the successful modern bicycle.

Selection 3 · Rosie Revere, Engineer

NARRATIVE POEM · ANDREA BEATY / DAVID ROBERTS · 8 QUESTIONS

Q17 RL.3.2

What is "Rosie Revere, Engineer" mostly about?

- ☐ **A.** How to keep snakes away with cheese spray.
- ☐ **B.** How airplanes are built in a factory.
- ☐ **C.** A shy girl who learns that failing is part of becoming a great inventor.
- ☐ **D.** How to win first prize at a science fair.

Q18 RL.3.3

Why does Rosie hide her dreams after she makes the cheese hat for Uncle Fred?

- ☐ **A.** She runs out of parts to build with.
- ☐ **B.** Her teacher tells her to stop.
- ☐ **C.** She moves to a new town.
- ☐ **D.** Uncle Fred laughs at her invention, and she feels embarrassed.

Q19 RL.3.3

How does Great-Great-Aunt Rose change what Rosie does?

- ☐ A. Her wish to fly inspires Rosie to start inventing again.
- ☐ B. She tells Rosie that inventing is a waste of time.
- ☐ C. She builds the cheese-copter for Rosie herself.
- ☐ D. She takes Rosie to live in the attic.

Q20 RL.3.3

What happens when Rosie tests her heli-o-cheese-copter?

- ☐ A. It flies away and never comes back.
- ☐ B. It flies for a moment, whirls around, and then crashes to the ground.
- ☐ C. It refuses to move at all.
- ☐ D. It turns into a real airplane.

Q21 RL.3.6

Aunt Rose says the crashed copter was "the perfect first try." What does she mean?

- ☐ A. The copter should never have been built.
- ☐ B. Rosie must build the very same copter again.
- ☐ C. The copter did fly before it crashed, so the try was a real success to learn from.
- ☐ D. Aunt Rose was only joking and did not mean it.

Q22 RL.3.5

How is this selection written?

- ☐ A. As a list of facts about machines.
- ☐ B. As a letter from Rosie to her aunt.
- ☐ C. As a play with parts for actors.
- ☐ D. As a narrative poem that tells a story in rhyming stanzas.

Q23 RL.3.3

By the end, how has Rosie changed?

- ☐ A. She keeps inventing and even teaches her classmates to cheer for their "perfect failures."
- ☐ B. She decides to stop inventing forever.
- ☐ C. She moves away from Blue River Creek.
- ☐ D. She gives all her tools to Uncle Fred.

Q24 RL.3.2 TRAP

What is the MAIN message, or theme, of "Rosie Revere, Engineer"?

- ☐ A. Rosie built a hat for Uncle Fred out of a fan and cheddar cheese spray.
- ☐ B. Failure is part of inventing, and the only true failure is quitting.
- ☐ C. Great-Great-Aunt Rose once built airplanes a long time ago.
- ☐ D. Rosie sat shyly in Lila Greer's classroom at Blue River Creek.

Selection 4 · Edison's Best Invention

OPINION TEXT · 8 QUESTIONS

Q25 RI.3.2

What is the author's opinion in this text?

- ☐ A. Of all Edison's inventions, the light bulb was his best.
- ☐ B. The phonograph was a waste of time.
- ☐ C. Edison should not be called an inventor.
- ☐ D. The kinetograph was Edison's only good idea.

Q26 RI.3.8

What two ways does the author use to compare Edison's inventions?

- ☐ A. How heavy each one is and how much it costs.
- ☐ B. How many new inventions each one inspired and how much each changed the way we live.
- ☐ C. What color each one is and how big it is.
- ☐ D. Who invented it first and who copied it.

Q27 RI.3.8

Which detail does the author use to support the claim about the light bulb?

- ☐ A. The light bulb was made of glass.
- ☐ B. Edison was partially deaf.
- ☐ C. The light bulb's power grid let people use electricity for refrigerators, washing machines, and computers.
- ☐ D. Edison named his inventions using Greek roots.

Q28 RI.3.6

Why does the author quote Gene Barretta, the writer of *Timeless Thomas*?

- ☐ A. To show that the author wrote *Timeless Thomas*.
- ☐ B. To prove that Barretta disagrees with the author.
- ☐ C. To explain how to build a light bulb.
- ☐ D. To use facts from a book as evidence that supports the author's opinion.

Q29 RI.3.1

According to the author, what was life like before the light bulb?

- ☐ **A.** People used candles, oil lamps, or gas lights, and burning them caused many deadly fires.
- ☐ **B.** People already had bright, safe electric lights.
- ☐ **C.** No one ever needed light after dark.
- ☐ **D.** Every street was lit by floodlights.

Q30 RI.3.3

According to the text, what did the phonograph lead to?

- ☐ **A.** Power grids that light whole cities.
- ☐ **B.** Vinyl records, compact discs, and digital music.
- ☐ **C.** The first motion picture camera.
- ☐ **D.** The first electric car.

Q31 RI.3.5

How does the author organize the part that compares the three inventions?

- ☐ **A.** By telling a made-up story about each one.
- ☐ **B.** By listing them in the order they were sold.
- ☐ **C.** By comparing and contrasting how many inventions each one inspired and how much each changed our lives.
- ☐ **D.** By writing a poem about each invention.

Q32 RI.3.2 **TRAP**

What is the MAIN idea of "Edison's Best Invention"?

- ☐ **A.** Edison built a movie studio and called it the Black Maria after a police wagon.
- ☐ **B.** Edison used Greek roots to name inventions like the kinetograph.
- ☐ **C.** The kinetograph was the first motion picture camera Edison built.
- ☐ **D.** A good invention both inspires new inventions and changes how we live, and the light bulb does both best.



Worksheet C — IB Critical Thinking

Module 8 · Imagine! Invent! · Grade 3

IB PYP Key Concepts · DOK 2 – 4

NAME _____ DATE _____

Directions. Use what you learned from each selection to think more deeply about what it takes to make a successful invention. Each question helps you grow as an IB learner.

Q1

Perseverance

DOK 3

In "**Timeless Thomas**," Edison did not stop when things went wrong. Show a **PROBLEM** or failure he faced, what he **DID** about it, and the **RESULT** for the world.

PROBLEM → ACTION → RESULT · THOMAS EDISON

A PROBLEM OR FAILURE

WHAT EDISON DID

THE RESULT

How did never giving up help Edison make successful inventions? Explain.

Q2

Change

DOK 3

In "**A Bumpy Ride**," the bicycle CHANGED step by step over many years. Trace THREE steps in that change, then explain how each inventor built on the one before.

HOW THE BICYCLE CHANGED · A BUMPY RIDE

STEP 1 · EARLY BIKE

what was the first machine like?

STEP 2 · AN IMPROVEMENT

what did a later inventor add?

STEP 3 · MODERN BIKE

how did it become safe and fast?

How did each inventor build on the work of the one before to make the bicycle better? Explain.

In "**Rosie Revere, Engineer**," Rosie learns what to do about FAILURE. Show how Rosie felt at first, what her aunt taught her, and how Rosie changed.

LEARNING FROM FAILURE · ROSIE REVERE

HOW ROSIE FELT AFTER THE CRASH

WHAT AUNT ROSE TAUGHT HER

What does Rosie learn about failure, and how does it change what she does? Why is this important for an inventor?

Q4

Innovation

DOK 3

Two selections show inventors at work in different ways. Compare HOW **Thomas Edison** in Timeless Thomas and **Rosie Revere** in Rosie Revere, Engineer each kept inventing.

TWO INVENTORS · TWO STORIES

THOMAS EDISON · A REAL INVENTOR

ROSIE REVERE · A STORY INVENTOR

What do both inventors show about what it takes to invent? (Hint: a real life vs. a poem, but the same idea about failure.)

Essential Question: What does it take to make a successful invention? Use ideas from TWO of the selections in this module to answer.

TWO-SELECTION SYNTHESIS

SELECTION 1 · TITLE

What does this selection show about making a successful invention (a brilliant idea, trying again after failure, building on others, hard work)?

SELECTION 2 · TITLE

What does this selection show about making a successful invention (a brilliant idea, trying again after failure, building on others, hard work)?

MY BIG IDEA



Worksheet D — MAP Practice

Module 8 · Imagine! Invent! · Grade 3

MAP RIT 195 → 215 · 4 Mini-Passages × 5 Questions

NAME _____ DATE _____

Directions. Read each short passage carefully. Then answer the 5 questions that follow. Each passage is NEW — not from the selections in this workbook.

Passage 1 · The Sticky Note

A scientist named Spencer was trying to make a very strong glue. Instead, he made a glue that was weak. It could stick to things, but it came off again easily. Spencer thought his work was a failure. He put the weak glue away. Years later, another worker named Art had a problem. The little papers that marked the pages in his book kept falling out. One day he remembered the weak glue. What if he put a bit of it on the back of a small paper? Now the paper could stick to a page and peel off without a tear. The weak glue was perfect for it. That is how the sticky note was born. Today people use sticky notes all over the world. The story shows that even a mistake can become a useful invention. Sometimes a brilliant idea comes from looking at a problem in a brand-new way.

152 WORDS · LEXILE 410–460L · RIT 195–201

Q1 RIT 195 Lexile 410L Key Detail

What was Spencer trying to make?

- ☐ A. A very strong glue.
- ☐ B. A new kind of book.
- ☐ C. A bright light bulb.
- ☐ D. A toy for children.

Q2 RIT 197 Lexile 420L Key Detail

What problem did Art have?

- ☐ A. His glue was too strong.
- ☐ B. The little papers that marked his pages kept falling out.
- ☐ C. His book was lost.
- ☐ D. His pen ran out of ink.

Q3 RIT 199 Lexile 440L Cause & Effect

What happened when Art put the weak glue on a small paper?

- ☐ A. The paper would not stick at all.
- ☐ B. The page tore in half.
- ☐ C. The paper could stick to a page and peel off without a tear.
- ☐ D. The glue dried up and was useless.

Q4 RIT 200 Lexile 450L Vocabulary

The passage says Spencer thought his work was a "failure." A **failure** is something that —

- ☐ A. works exactly as planned.
- ☐ B. is very expensive.
- ☐ C. is bright and shiny.
- ☐ D. does not work the way you wanted.

Q5 RIT 201 Lexile 460L Main Idea

What is the BIG IDEA of "The Sticky Note"?

- ☐ A. Glue is made by scientists.
- ☐ B. Even a mistake can become a useful invention when you see a problem in a new way.
- ☐ C. Books have many pages.
- ☐ D. Sticky notes come in many colors.

Passage 2 · The Better Pencil

Mia loved to draw, but she had a problem. When she made a mistake, she had to find an eraser. Her eraser was always rolling off the desk or hiding in her bag. One afternoon she lost her eraser three times. "There must be a better way," Mia thought. She looked at her pencil and then at the eraser. What if the two could stay together? Mia found a small metal ring and a bit of tape. Carefully, she fixed the eraser to the top of her pencil with the ring. Now the eraser was always right there when she needed it. She never had to search for it again. Her friends saw her clever pencil and wanted one too. Mia was proud. She had taken two things that already existed and joined them to solve a real problem. A good invention does not always have to be brand new. Sometimes it just makes everyday life a little easier.

159 WORDS · LEXILE 470–530L · RIT 202–207

Q6 RIT 202 Lexile 470L Key Detail

What problem did Mia have?

- ☐ A. Her pencil kept breaking.
- ☐ B. Her eraser was always rolling away or getting lost.
- ☐ C. She did not know how to draw.
- ☐ D. Her desk was too small.

Q7 RIT 204 Lexile 490L Key Detail

What did Mia do to solve her problem?

- ☐ A. She bought ten new erasers.
- ☐ B. She stopped drawing.
- ☐ C. She glued her desk to the floor.
- ☐ D. She fixed the eraser to the top of her pencil with a small ring.

Q8 RIT 205 Lexile 500L Cause & Effect

What happened after Mia joined the eraser to her pencil?

- ☐ A. The eraser was always there, and she never had to search for it again.
- ☐ B. The pencil stopped working.
- ☐ C. She lost the pencil too.
- ☐ D. Her friends told her to take it apart.

Q9 RIT 206 Lexile 510L Inference

When Mia's friends "wanted one too," it suggests that —

- ☐ A. the friends did not like Mia's idea.
- ☐ B. the pencil was broken.
- ☐ C. her invention was useful to other people, not just to her.
- ☐ D. Mia kept her idea a secret.

Q10 RIT 207 Lexile 530L Main Idea

What is the BIG IDEA of "The Better Pencil"?

- ☐ A. Erasers are made of rubber.
- ☐ B. A good invention can join two ordinary things to make everyday life easier.
- ☐ C. Drawing is a fun hobby.
- ☐ D. Desks should be kept clean.

Passage 3 · How the Bicycle Bell Rang

Long ago, riders on busy streets had a problem. People walking did not hear the bicycles coming. There were many small crashes. Riders shouted to warn people, but a shout was easy to miss in the noise of the city. An inventor wanted a sound that was louder and clearer than a voice. He thought about the small bells used on doors and shop counters. What if a bell could be placed right on the handlebar? The rider could press it with a thumb without letting go of the bike. The inventor built a little metal dome with a spring inside. When the rider pressed a lever, a tiny hammer struck the dome and made a bright ring. People could hear it from far away. Soon bicycle bells were everywhere, and the streets grew safer. The bell did not change the bicycle itself. It was a small idea, but it solved a real problem and helped many people stay safe.

160 WORDS · LEXILE 540–600L · RIT 208–212

Q11 RIT 208 Lexile 540L Key Detail

What was the problem on busy streets long ago?

- ☐ A. The bicycles were too slow.
- ☐ B. There were no streets at all.
- ☐ C. People walking did not hear the bicycles coming, so there were many crashes.
- ☐ D. The bells were too loud.

Q12 RIT 209 Lexile 560L Cause & Effect

Why did shouting not work well as a warning?

- ☐ A. A shout was easy to miss in the noise of the city.
- ☐ B. The riders could not speak.
- ☐ C. Shouting was against the rules.
- ☐ D. No one walked on the streets.

Q13 RIT 210 Lexile 570L Key Detail

How did the bicycle bell make its sound?

- ☐ A. A motor spun a wheel.
- ☐ B. The rider blew into a horn.
- ☐ C. The wind passed over a hole.
- ☐ D. A tiny hammer struck a metal dome when the rider pressed a lever.

Q14 RIT 211 Lexile 580L Inference

The bell could be pressed "with a thumb without letting go of the bike." This was helpful because —

- ☐ A. the rider wanted to ring it with a foot.
- ☐ B. the rider could warn people and still steer the bike safely.
- ☐ C. the bell was too heavy to lift.
- ☐ D. the rider did not need to steer.

Q15 RIT 212 Lexile 600L Main Idea

What is the BIG IDEA of "How the Bicycle Bell Rang"?

- ☐ A. Bells are made of metal.
- ☐ B. City streets are always noisy.
- ☐ C. Bicycles are faster than walking.
- ☐ D. A small invention can solve a real problem and help keep many people safe.

Passage 4 · The Boy Who Made Braille

Louis Braille became blind as a young boy. At his school, blind students could read only a few books. The letters were large and raised, and the books were heavy and hard to make. Louis wanted blind people to read and write as easily as anyone else. He learned about a secret code of raised dots that soldiers used to send messages in the dark. The code was too hard for most people, but it gave Louis a brilliant idea. He worked for years to make the code simpler. He created a system of six raised dots. Each pattern of dots stood for a letter. A blind reader could feel the dots with a fingertip and read quickly. A blind writer could press the dots onto paper. Louis was only fifteen when he finished his system. Today, blind people all over the world read and write with Braille. One determined young person changed millions of lives with a simple, clever idea.

161 WORDS · LEXILE 610–680L · RIT 213–215

Q16 RIT 213 Lexile 610L Key Detail

What problem did blind students have at Louis's school?

- ☐ A. They could read only a few books, which were heavy and hard to make.
- ☐ B. They had no school at all.
- ☐ C. They had too many easy books.
- ☐ D. They did not want to read.

Q17 RIT 214 Lexile 630L Key Detail

Where did Louis get the idea for his system?

- ☐ A. From a book about light bulbs.
- ☐ B. From a teacher's drawing.
- ☐ C. From a secret code of raised dots that soldiers used in the dark.
- ☐ D. From a song he heard at school.

Q18 RIT 214 Lexile 650L Vocabulary

The passage says Louis had a "brilliant idea." A **brilliant** idea is one that is —

- ☐ A. dull and not helpful.
- ☐ B. very clever and smart.
- ☐ C. copied from a friend.
- ☐ D. impossible to use.

Q19 RIT 215 Lexile 670L Cause & Effect

What can a blind reader do because of Louis's system?

- ☐ A. Hear the words spoken aloud.
- ☐ B. Watch the letters light up.
- ☐ C. Smell the ink on the page.
- ☐ D. Feel the raised dots with a fingertip and read quickly.

Q20 RIT 215 Lexile 680L Main Idea

What is the BIG IDEA of "The Boy Who Made Braille"?

- ☐ A. Soldiers send messages in the dark.
- ☐ B. Schools have many books.
- ☐ C. One determined person, with a simple clever idea, can change millions of lives.
- ☐ D. Dots can be pressed onto paper.